How our atmosphere obscures planets: a model of the effect of telluric contamination on solar emission lines

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I. Abstract

Ground-based telescopes continuously observe solar spectra to extract precise radial velocity measurements, a technique essential to exoplanet detection and stellar characterization. However, before this light reaches a telescope's detector, it must pass through Earth's atmosphere, where primarily water vapor (H2O) and carbon dioxide (CO2) selectively absorb light at specific wavelengths. These telluric lines frequently overlap with spectral lines, contaminating the observed spectrum and introducing systematic errors into measurements. This project models telluric contamination using the Voigt profile, a convolution of Gaussian and Lorentzian line shapes that captures both the Doppler (thermal/turbulent) and pressure-broadening components characteristic of atmospheric absorption lines. As the Voigt profile will be modeled to represent a spectral line, telluric lines will also be introduced at varying locations where they become superimposed at varying wavelength offsets relative to the spectral line center, simulating different degrees of contamination. By systematically varying the position of telluric contamination across the spectral line profile, this study quantifies how the location of overlap affects the accuracy of derived radial velocity shifts. The results aim to create a visual representation of how differing telluric contamination on a spectral line can create greater or lesser impacts in data, providing insight into how telluric line position and severity should be accounted for in radial velocity shift pipelines used for solar and stellar observations.

II. Background

A. The Doppler Effect

The Doppler effect is known as the change in observed frequency or wavelength of a wave relative to an observer. If the source approaches the observer, the waves are compressed. As the source moves away from the observer, the waves stretch out. For light waves, this motion of a source creates a change in color. Blueshift; if the source is moving towards the observer, the waves are compressed and are at a higher wavelength. Redshift; if the source is moving away from the observer, the waves are stretched and at a lower wavelength.

Measuring a stars Doppler shift gives us the ability to determine it's radial velocity. As we receive the wavelengths that a star gives off, not to get confused, the Doppler shift of a star is measured as the change in the wavelength of light. This is the *physical phenomenon* and our *observable data.* The radial velocity however, is the actual speed of a stars wavelength that is either moving towards or away from us, the observer.

We can use the Doppler shift formula, rearranged, to isolate different variables that we want. In this case, we would isolate the radial velocity variable to calculate it's value. Where $\Delta\lambda =$ the amount the wavelength has shifted. $\lambda_0 =$ the original unshifted wavelength. $V_r =$ the radial velocity. And $c =$ the speed of light.

$$\frac{\Delta\lambda}{\lambda_0} = \frac{V_r}{c}$$

When a planet orbits a star, it creates a gravitational tug that causes that star to "wobble" back and forth. This wobble is because the planet does not perfectly orbit its stars center, but rather both the star and the planet have a 'shared center of mass', known as the *barycenter.*

Measuring this periodic shift in a stars spectrum, we are able to calculate the planets mass and other orbital characteristics. This is useful to also detect hidden exoplanets orbiting distant stars.

B. The Voigt profile

The Voigt profile is a mathematical curve that describes the shape of spectral lines. It is most commonly used to fit spectral graphs, to find a center point. The Voigt profile itself is created by convolving two different curves, such as the *Gaussian curve* and the *Lorentzian curve.* The Voigt profile is important because it provides a real-world representation of absorption and emission lines. I will be using Pythons "scipy.special.voigt–profile" to model and use it's graph.

III. Methods

The Voigt profile will be used to model a spectral line, instead of using data from a real spectral line we will be generating modeled data ourselves. Graphing in the Voigt profile along with two "noise" representing telluric lines, this models spectral data that we could have received from a telescope, of course, if the spectral line was perfect. In this case since we are dealing with a very smooth representing spectral line, we can amplify the noise. Calculating the center point with the "Full-Width-Half-Maximum" method, we repeat this process of finding the center line as we shift the graph along the axes, moving it through the telluric lines. This is done because light waves are moving as they approach us. To model the "Doppler Effect" on the produced data, we make the Voigt profile "wobble" slightly back and forth as it shifts along the axes. The center line points then are used to calculate the radial velocity shift. Once graphed, we compare the radial velocity shift results where a Voigt profile consisted of telluric lines, and a Voigt profile without any telluric contamination. *Note: The first two shown graphs are *not to scale* and serves the purpose for visual representation of the Full Width Half Maximum method only.

IV. Results

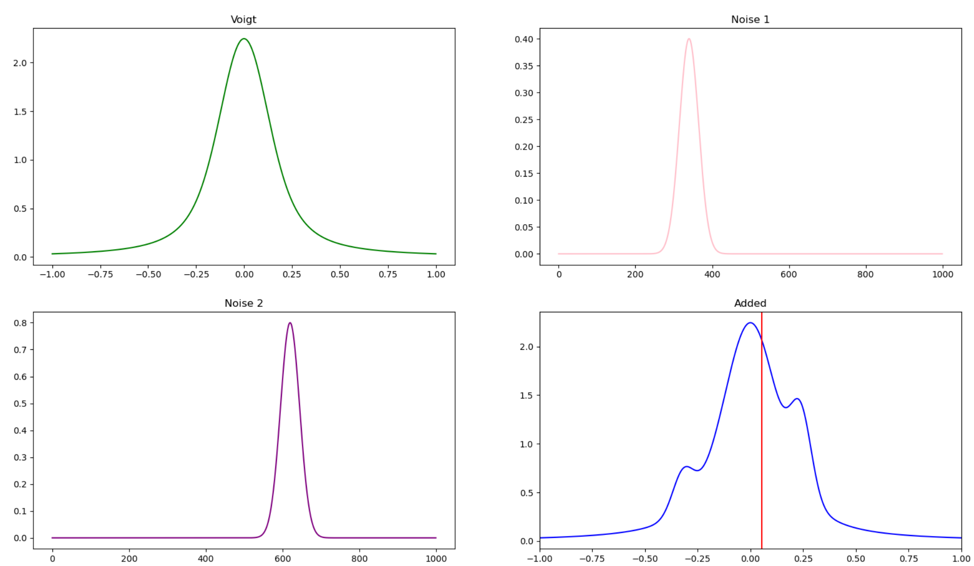

FIG. 1. Representation of Voigt profile with and without telluric contamination, along with a visual of where the center point is on the contaminated Voigt profile.

After figuring out the "Full With Half Maximum" method, we are now able to shift the Voigt profile with contamination along the axis. This is done because we want to find the center point over time, which models how a telescope would be taking snapshots and collecting data every minute or so. (Which is relatively slow)

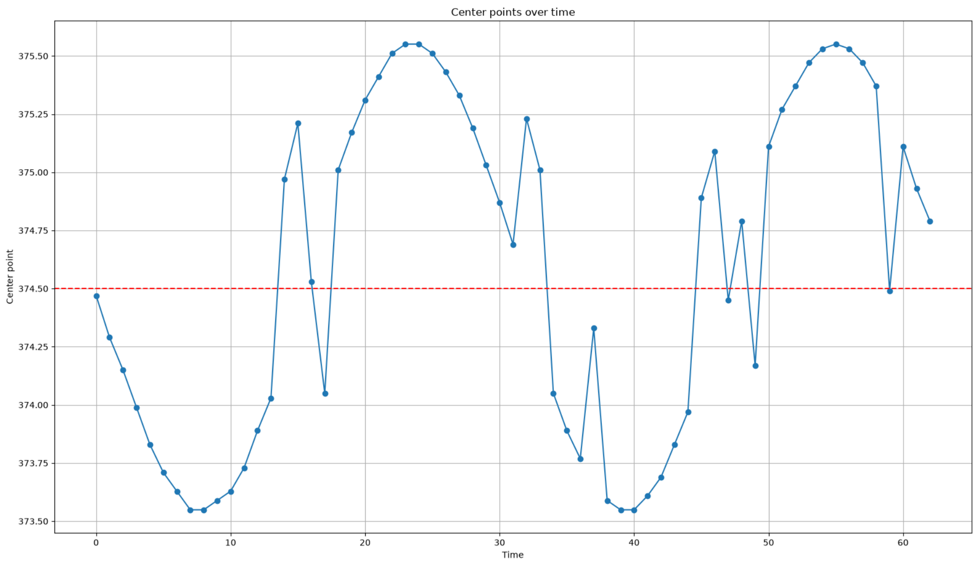

FIG. 2. This is a graph of the Voigt profile with 3 telluric lines, one shifted left, two shifted right with each one having a different amplitude.

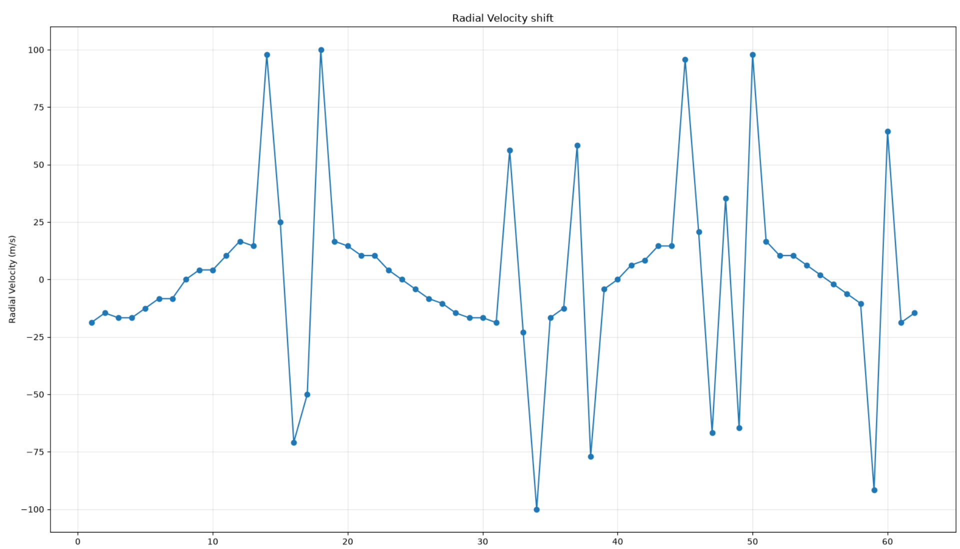

FIG. 3. This is the graph of the calculated Radial Velocity Shift of the center points in FIG.2.

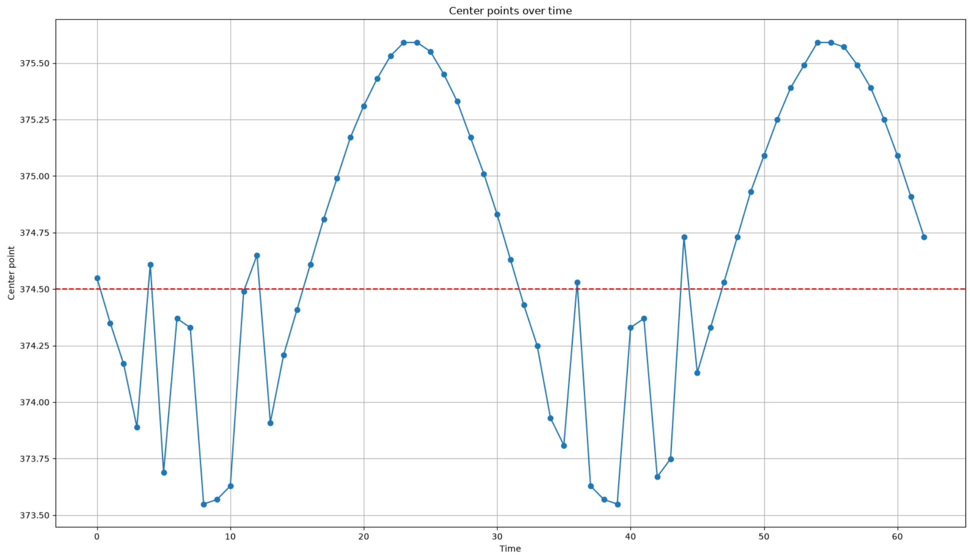

FIG. 4. Another example of the Voigt profile with differed telluric lines, the amplitude and location of the telluric lines has been slightly altered.

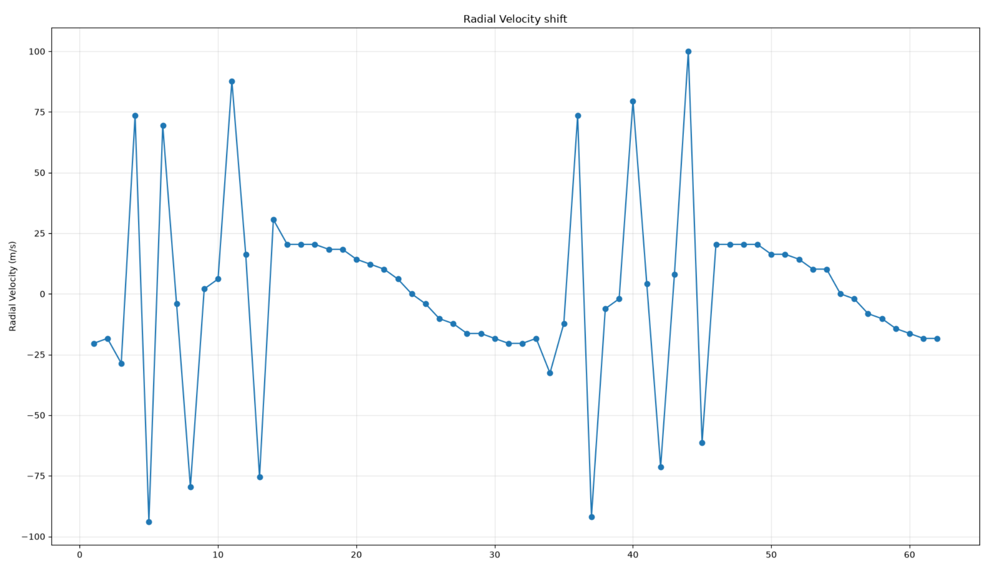

FIG. 5. The graph of the Radial Velocity Shift of FIG.4.

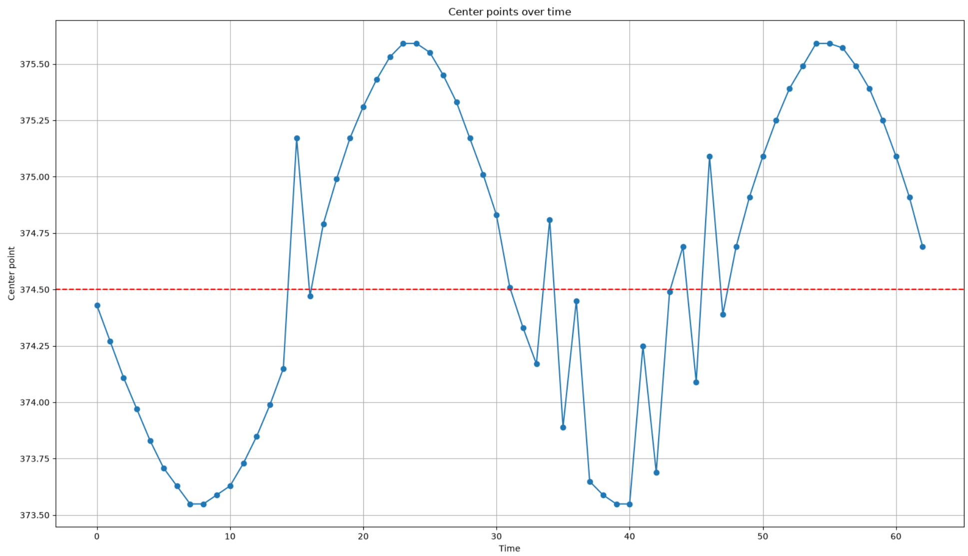

FIG. 6. This graph shows the Voigt profile again with 3 telluric lines, however the intensity of the telluric lines has been decreased, such as the amplitude. The location of the telluric lines also shifted a bit.

V. Discussion

The results shown in FIG. 1 through FIG. 5 demonstrate that the position of telluric contamination relative to the spectral emission line center has a substantial effect on the accuracy of the derived radial velocity shift. When the telluric contamination overlapped directly; or closely with the peak of our modeled spectral line, the calculated center point was displaced most significantly, causing the output graph to have varied intense "jumps". This is consistent with the expectation that the centering finding method–the *Full-Width-Half-Maximum* method is most sensitive to distortions near the peak of the profile, where small changes in the summed intensity have the greatest effect on the location of the calculated maximum. In contrast, contamination located further into the wings of the Voigt profile produced a comparatively smaller radial velocity shift, with lesser jumps in both the center point graph over time and radial velocity shift over time. This is because the wings of our modeled spectral line contribute less to the overall shape near the peak, so the FWHM-derived center point is less sensitive to distortions in that region. Fully blended overlaps, where the telluric and spectral emission line were nearly indistinguishable, produced a great amount of change in where the center point was located when graphed, which shows that the degree of separation between the two line centers plays an important role in how easily contamination can be identified and corrected for. Comparing the calculated radial velocity shifts corresponding to the center points, we notice that as the telluric contamination is more intense, the radial velocity shift graph tends to create more "jumps", which we observe is where the contamination is most significant. This suggests that even relatively small telluric contamination, if located near the spectral line center, could impact radial velocity shift measurements in real observational data greatly.

These results highlight the importance of telluric line identification and removal, particularly for spectral lines that rest wavelengths happen to fall near strong atmospheric absorption molecules like H2O and CO2. But also the affects of how telluric contamination at different locations can have the greater or lesser impact on the shifting center points and calculated radial velocity shift of a spectra line.

VI. Conclusion

In this project the effect of telluric contamination on radial velocity measurements was modeled by using a representation of the shape of a spectral line, the Voigt profile. By systematically varying the position of a simulated telluric line across the Voigt profile, we were able to observe that the location of contamination significantly affects the accuracy of the derived radial velocity shift, with contamination near the line peak producing the greatest change and error. These results demonstrate that telluric absorption is not a uniform source of error, but rather one whose impact depends strongly on the specific wavelength region it overlaps with. For future reference; this model could be extended to include real telluric absorption spectra rather than simplified Voigt profile approximations, and could explore how contamination severity, in addition to position, further affects radial velocity shift precision, while also comparing to real data.